Adaptive Query Selection (AQS)

L2 Class Confidence

$s_i = \max_c \sigma(S2_{i,c})$

0.21	0.83	0.74	0.36	0.92	0.68	0.27	0.89

Learned Threshold

$\tau = \sigma(\theta)$

T = 0.10 (fixed)

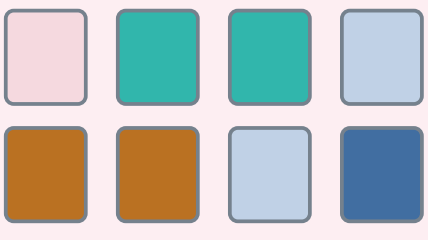
Adaptive Gating

$g_i = 1[s_i \geq \tau]$

0	1	1	0	1	1	0	1

gate g

Query Features Q2



Query Pooling

$c = \Sigma(g_i\, q_i) / \max(\Sigma g_i, 1)$

context c

query Q2

Context Fusion

$\Delta Q = \mathrm{MLP}([\mathrm{LN}(Q2); c])$

identity Q2

Gated Residual

$\tanh(\rho) \cdot g \cdot \Delta Q$

+

Refined Queries Q2′ (300 → 300)



To Decoder L3

Learned τ, ρ · Hard forward gate · STE during training